

Purna Sai

Enhanced Seq2Seq Autoencoder via Contrastive Learning for Abstractive Text Summarization

```
pip install transformers
```

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Looking in indexes: https://pypi.org/simple, https://us-python.pkg.dev/colab-wheels/public/simple/
Collecting transformers
  Downloading transformers-4.26.1-py3-none-any.whl (6.3 MB)
    6.3/6.3 MB 40.4 MB/s eta 0:00:00
Requirement already satisfied: filelock in /usr/local/lib/python3.8/dist-packages (from transformers) (3.9.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.8/dist-packages (from transformers) (4.64.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.8/dist-packages (from transformers) (6.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.8/dist-packages (from transformers) (1.22.4)
Requirement already satisfied: requests in /usr/local/lib/python3.8/dist-packages (from transformers) (2.25.1)
Collecting tokenizers!=0.11.3,<0.14,>=0.11.1
  Downloading tokenizers-0.13.2-cp38-cp38-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (7.6 MB)
    7.6/7.6 MB 86.6 MB/s eta 0:00:00
Collecting huggingface-hub<1.0,>=0.11.0
  Downloading huggingface_hub-0.12.1-py3-none-any.whl (190 kB)
    190.3/190.3 KB 19.7 MB/s eta 0:00:00
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.8/dist-packages (from transformers) (23.0)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.8/dist-packages (from transformers) (2022.6.2)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.8/dist-packages (from huggingface-hub<1.0,>=0.11.0) (4.0.0)
Requirement already satisfied: chardet<5,>=3.0.2 in /usr/local/lib/python3.8/dist-packages (from requests->transformers) (4.0.0)
Requirement already satisfied: urllib3<1.27,>=1.21.1 in /usr/local/lib/python3.8/dist-packages (from requests->transformers) (1.24.0)
Requirement already satisfied: idna<3,>=2.5 in /usr/local/lib/python3.8/dist-packages (from requests->transformers) (2.10)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.8/dist-packages (from requests->transformers) (2022.12.7)
Installing collected packages: tokenizers, huggingface-hub, transformers
Successfully installed huggingface-hub-0.12.1 tokenizers-0.13.2 transformers-4.26.1
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```
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
```

```
class Seq2SeqAutoencoder(nn.Module):
    def __init__(self, input_size, hidden_size, output_size):
        super(Seq2SeqAutoencoder, self).__init__()
        self.encoder = nn.LSTM(input_size, hidden_size, batch_first=True)
        self.decoder = nn.LSTM(hidden_size, output_size, batch_first=True)
        self.linear = nn.Linear(output_size, input_size)

    def forward(self, x):
        _, hidden = self.encoder(x)
        output, _ = self.decoder(hidden)
        x_hat = self.linear(output)
        return x_hat.squeeze()

class ContrastiveLoss(nn.Module):
    def __init__(self, margin):
        super(ContrastiveLoss, self).__init__()
        self.margin = margin

    def forward(self, anchor, positive, negative):
        distance1 = torch.norm(anchor - positive, dim=1)
        distance2 = torch.norm(anchor - negative, dim=1)
        loss = torch.mean(torch.clamp(distance1 - distance2 + self.margin, min=0))
        return loss

class TextDataset(Dataset):
    def __init__(self, texts, summaries):
        self.texts = texts
        self.summaries = summaries

    def __len__(self):
        return len(self.texts)

    def __getitem__(self, index):
        text = self.texts[index]
        summary = self.summaries[index]
        return text, summary
```

```

class TextCollator(object):
    def __init__(self, tokenizer):
        self.tokenizer = tokenizer

    def __call__(self, batch):
        texts, summaries = zip(*batch)
        texts = [self.tokenizer.encode(text, max_length=512, truncation=True, padding='max_length') for text in texts]
        summaries = [self.tokenizer.encode(summary, max_length=64, truncation=True, padding='max_length') for summary in summaries]
        return torch.tensor(texts), torch.tensor(summaries)

def train(model, dataloader, optimizer, criterion):
    model.train()
    running_loss = 0.0
    for texts, summaries in dataloader:
        optimizer.zero_grad()
        texts, summaries = texts.to(device), summaries.to(device)
        encoded = model.encoder(texts)[1][0]
        decoded = model.decoder(encoded)[0]
        positive = decoded[:, :-1, :]
        negative = torch.cat([decoded[:, 1:, :], decoded[:, 0:1, :]], dim=1)
        loss = criterion(encoded, positive, negative)
        loss.backward()
        optimizer.step()
        running_loss += loss.item() * texts.size(0)
    epoch_loss = running_loss / len(dataloader.dataset)
    return epoch_loss

def evaluate(model, dataloader, criterion):
    model.eval()
    running_loss = 0.0
    with torch.no_grad():
        for texts, summaries in dataloader:
            texts, summaries = texts.to(device), summaries.to(device)
            encoded = model.encoder(texts)[1][0]
            decoded = model.decoder(encoded)[0]
            positive = decoded[:, :-1, :]
            negative = torch.cat([decoded[:, 1:, :], decoded[:, 0:1, :]], dim=1)
            loss = criterion(encoded, positive, negative)
            running_loss += loss.item() * texts.size(0)
    epoch_loss = running_loss / len(dataloader.dataset)
    return epoch_loss

```

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